Continuity and Differentiability: Questions and Full Solutions

Question 01

English

Let $p(x)$ be a cubic polynomial with real coefficients and leading coefficient 1 such that $p(1) = 0$. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = |p(x)|$. It is given that f is not differentiable at exactly one point of $\mathbb{R}$, namely at $x = 3$. Then $f'(0)$ is equal to

(A) 7
(B) -3
(C) -7
(D) 3

Only one option is correct.

हिन्दी

माना कि $p(x)$ वास्तविक गुणांकों वाला एक त्रिघात बहुपद है जिसका अग्रग गुणांक 1 है तथा $p(1) = 0$ है। माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = |p(x)|$ द्वारा परिभाषित है। दिया गया है कि f, $\mathbb{R}$ के ठीक एक बिन्दु, अर्थात $x = 3$ पर, अवकलनीय नहीं है। तब $f'(0)$ बराबर है

(A) 7
(B) -3
(C) -7
(D) 3

केवल एक विकल्प सही है।

Solution

Away from the real roots of p, $f = |p|$ equals p or $-p$ on a neighbourhood, so it is differentiable there.

At a real root r: if $p'(r) \neq 0$, then $p(x) \approx p'(r)(x - r)$ changes sign at r, and f has left and right derivatives $-|p'(r)|$ and $|p'(r)|$, which differ; so f is not differentiable at r. If $p'(r) = 0$, then $\dfrac{|p(x)|}{|x - r|} \to 0$, so $f'(r) = 0$ exists.

Hence f is non-differentiable exactly at the simple real roots of p.

Since f is not differentiable at $x = 3$, 3 is a simple root of p. Since $p(1) = 0$ and f is differentiable at $x = 1$, the root 1 is not simple, so $(x - 1)^2$ divides $p(x)$.

With leading coefficient 1 and roots $1, 1, 3$: $p(x) = (x - 1)^2(x - 3)$. Its only simple root is 3, so the condition holds.

$p'(x) = 2(x-1)(x-3)+(x-1)^2 = (x-1)(2x-6+x-1) = (x-1)(3x-7)$, so $p'(0) = (-1)(-7) = 7$.

$p(0) = (1)(-3) = -3 < 0$; by continuity $p < 0$ near 0, so $f(x) = -p(x)$ there.

Therefore $f'(0) = -p'(0) = -7$.

Correct Option: (C)

Question 02

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \{x\} \sin\left(\frac{\pi x}{2}\right) \sin\left(\frac{\pi x}{3}\right),$$

where $\{t\} = t - [t]$ and $[t]$ denotes the greatest integer less than or equal to t. Let α be the number of points of the interval $(0, 50)$ at which f is discontinuous, and let β be the number of points of the interval $(0, 50)$ at which f is not differentiable. Then the value of $\alpha + \beta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$ इस प्रकार परिभाषित है

$$f(x) = \{x\} \sin\left(\frac{\pi x}{2}\right) \sin\left(\frac{\pi x}{3}\right),$$

जहाँ $\{t\} = t - [t]$ है तथा $[t]$, t से कम या उसके बराबर महत्तम पूर्णांक को दर्शाता है। माना कि α अन्तराल $(0, 50)$ के उन बिन्दुओं की संख्या है जिन पर f असतत है, तथा β अन्तराल $(0, 50)$ के उन बिन्दुओं की संख्या है जिन पर f अवकलनीय नहीं है। तब $\alpha + \beta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Write $p(x) = \sin\left(\frac{\pi x}{2}\right) \sin\left(\frac{\pi x}{3}\right)$, a differentiable function on $\mathbb{R}$, so $f(x) = \{x\}p(x)$.

On each open interval $(n, n+1)$, $\{x\} = x - n$ is a polynomial, so f is differentiable there. Only the integers $n = 1, 2, \ldots, 49$ need checking.

Fix an integer n. For $x \in (n-1, n)$, $f(x) = (x - n + 1)p(x)$; for $x \in [n, n+1)$, $f(x) = (x - n)p(x)$.

Continuity: $\lim_{x \to n^-} f(x) = p(n)$ and $\lim_{x \to n^+} f(x) = 0 = f(n)$. Hence f is continuous at n iff $p(n) = 0$.

$p(n) = 0$ iff $\sin\frac{\pi n}{2} = 0$ or $\sin\frac{\pi n}{3} = 0$, i.e. iff n is divisible by 2 or by 3.

Differentiability at such n: left derivative $= [(x - n + 1)p(x)]'_{x=n} = p(n) + p'(n) = p'(n)$; right derivative $= [(x - n)p(x)]'_{x=n} = p(n) + 0 = 0$. So f is differentiable at n iff $p'(n) = 0$.

$p'(n) = \frac{\pi}{2} \cos\frac{\pi n}{2} \sin\frac{\pi n}{3} + \frac{\pi}{3} \sin\frac{\pi n}{2} \cos\frac{\pi n}{3}$. If exactly one of the sines vanishes, the surviving term is non-zero (the cosine of a zero of sine is ± 1). If both vanish ($6 \mid n$), $p'(n) = 0$. So f is differentiable at n iff $6 \mid n$.

Counting in $\{1, \ldots, 49\}$: multiples of 2: 24; of 3: 16; of 6: 8. Divisible by 2 or 3: $24 + 16 - 8 = 32$.

α = integers not divisible by 2 or $3 = 49 - 32 = 17$.

β = discontinuity points (17) + continuous but non-differentiable points $(32 - 8 = 24) = 41$.

Hence $\alpha + \beta = 17 + 41 = 58$.

Final Answer: 58

Question 03

English

Let $a \in \mathbb{R}$ and let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} a - 2x - x^3, & x \leq 0, \\ 2 - 5\log_e(1 + x), & x > 0. \end{cases}$$

Suppose that f is continuous on $\mathbb{R}$. Then f is a bijection; let $g : \mathbb{R} \to \mathbb{R}$ be its inverse function. Let

$$L = \lim_{h \to 0^-} \frac{g(2+h) - g(2)}{h}, \qquad R = \lim_{h \to 0^+} \frac{g(2+h) - g(2)}{h}.$$

Then the value of $R - L$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $a \in \mathbb{R}$ तथा $f : \mathbb{R} \to \mathbb{R}$ इस प्रकार परिभाषित है

$$f(x) = \begin{cases} a - 2x - x^3, & x \leq 0, \\ 2 - 5\log_e(1+x), & x > 0. \end{cases}$$

माना कि f, $\mathbb{R}$ पर सतत है। तब f एक एकैकी आच्छादी फलन है; माना कि $g : \mathbb{R} \to \mathbb{R}$ इसका प्रतिलोम फलन है। माना कि

$$L = \lim_{h \to 0^-} \frac{g(2+h) - g(2)}{h}, \qquad R = \lim_{h \to 0^+} \frac{g(2+h) - g(2)}{h}.$$

तब $R - L$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Continuity at $x = 0$: $f(0) = a$ and $\lim_{x \to 0^+} f(x) = 2 - 5\log_e 1 = 2$, so $a = 2$.

For $x < 0$: $f'(x) = -2 - 3x^2 < 0$; for $x > 0$: $f'(x) = -\dfrac{5}{1+x} < 0$. With continuity at 0, f is strictly decreasing on $\mathbb{R}$. As $x \to -\infty$, $f(x) \to +\infty$ and as $x \to \infty$, $f(x) \to -\infty$, so f is a bijection of $\mathbb{R}$.

$f(0) = 2$, so $g(2) = 0$. One-sided derivatives of f at 0: $f'(0^-) = -2$ (from $a - 2x - x^3$) and $f'(0^+) = -5$ (from $2 - 5\log_e(1+x)$).

For $h < 0$: $2 + h < 2 = f(0)$, and since f is decreasing, $k = g(2+h) > 0$ with $k \to 0^+$ as $h \to 0^-$ (continuity of the inverse of a continuous monotonic function). Then

$$\frac{g(2+h) - g(2)}{h} = \frac{k}{f(k) - f(0)} \to \frac{1}{f'(0^+)} = -\frac{1}{5}.$$

So $L = -\dfrac{1}{5}$.

For $h > 0$: $k = g(2+h) < 0$ with $k \to 0^-$, and the quotient tends to $\dfrac{1}{f'(0^-)} = -\dfrac{1}{2}$. So $R = -\dfrac{1}{2}$.

Therefore $R - L = -\dfrac{1}{2} + \dfrac{1}{5} = -\dfrac{3}{10} = -0.30$.

Final Answer: -0.30

Question 04

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \max\left\{\{x\}, \dfrac{|x|}{4}\right\}$, where $\{t\} = t - [t]$ and $[t]$ denotes the greatest integer less than or equal to t. Then the sum of all the points of the interval $(-4, 4)$ at which f is not differentiable is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \max\left\{\{x\}, \frac{|x|}{4}\right\}$ द्वारा परिभाषित है, जहाँ $\{t\} = t - [t]$ है तथा $[t]$, t से कम या उसके बराबर महत्तम पूर्णांक को दर्शाता है। तब अन्तराल $(-4, 4)$ के उन सभी बिन्दुओं का योग, जिन पर f अवकलनीय नहीं है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Write $u(x) = \{x\}$ and $v(x) = \dfrac{|x|}{4}$. On $(-4, 4)$, $0 \le v < 1$.

Integer points $n \in \{-3, \ldots, 3\}$: $\lim_{x \to n^-} u(x) = 1$ while $v(n) = \frac{|n|}{4} \le \frac{3}{4} < 1$, so $\lim_{x \to n^-} f(x) = 1$. But $f(n) = \max\left\{0, \frac{|n|}{4}\right\} = \frac{|n|}{4} \ne 1$. So f is discontinuous, and hence not differentiable, at each of these 7 integers (including 0).

Non-integer points: f is the maximum of two functions that are differentiable away from integers and 0; f fails to be differentiable only where $u = v$ with different slopes.

For $x \in (n, n+1)$ with $n \ge 0$: $u = x - n$ (slope 1), $v = \frac{x}{4}$ (slope $\frac{1}{4}$). $u = v \iff x = \frac{4n}{3}$. This lies in $(n, n+1)$ iff $0 < \frac{n}{3} < 1$, i.e. $n = 1, 2$: $x = \frac{4}{3}, \frac{8}{3}$. (For $n = 3$, $x = 4$ is outside $(3, 4)$.)

For $x \in (n, n+1)$ with $n \le -1$: $u = x - n$ (slope 1), $v = -\frac{x}{4}$ (slope $-\frac{1}{4}$). $u = v \iff x = \frac{4n}{5}$, which lies in $(n, n+1)$ iff $-\frac{n}{5} < 1$, true for $n = -4, -3, -2, -1$: $x = -\frac{16}{5}, -\frac{12}{5}, -\frac{8}{5}, -\frac{4}{5}$.

At each such crossing the slopes differ, so there is a corner.

Sum of the integer points: $-3 - 2 - 1 + 0 + 1 + 2 + 3 = 0$.

Sum of the crossings: $\frac{4}{3} + \frac{8}{3} - \frac{16+12+8+4}{5} = 4 - 8 = -4$.

Total sum $= -4$ (13 points in all).

Final Answer: -4

Question 05

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(t) = \begin{cases} 2t - t^2, & t \le 2, \\ \dfrac{3}{2}(t - 2), & t > 2, \end{cases}$$

and let $g : \mathbb{R} \to \mathbb{R}$ be defined by $g(x) = \max\{f(t) : x - 1 \le t \le x\}$. Let S be the set of all points of the interval $(0, 5)$ at which g is not differentiable. Then the sum of all the elements of S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$ इस प्रकार परिभाषित है

$$f(t) = \begin{cases} 2t - t^2, & t \le 2, \\ \dfrac{3}{2}(t - 2), & t > 2, \end{cases}$$

तथा माना कि $g : \mathbb{R} \to \mathbb{R}$, $g(x) = \max\{f(t) : x - 1 \le t \le x\}$ द्वारा परिभाषित है। माना कि S अन्तराल $(0, 5)$ के उन सभी बिन्दुओं का समुच्चय है जिन पर g अवकलनीय नहीं है। तब S के सभी अवयवों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Shape of f: for $t \le 2$, $f(t) = 1 - (t-1)^2$, increasing on $(-\infty, 1]$, maximum 1 at $t = 1$, decreasing on $[1, 2]$ with $f(2) = 0$. For $t > 2$, $f(t) = \frac{3}{2}(t-2)$ is increasing and f is continuous at 2.

Case $x \le 1$: $[x-1, x] \subset (-\infty, 1]$ where f increases, so $g(x) = f(x) = 2x - x^2$ and $g'(x) = 2 - 2x$.

Case $1 \le x \le 2$: the window contains $t = 1$, the global maximum point of f on $(-\infty, 2]$, and $[x-1, x] \subset (-\infty, 2]$. So $g(x) = 1$.

Case $2 \le x \le 3$: on $[x-1, 2] \subset [1, 2]$, f decreases, so its maximum is $f(x-1) = 1 - (x-2)^2$; on $(2, x]$ the maximum is $f(x) = \frac{3}{2}(x-2)$. Hence $g(x) = \max\left\{1 - (x-2)^2, \frac{3}{2}(x-2)\right\}$.

With $u = x - 2 \in [0, 1]$: $1 - u^2 \ge \frac{3}{2}u \iff 2u^2 + 3u - 2 \le 0 \iff (2u-1)(u+2) \le 0 \iff u \le \frac{1}{2}$. So $g = 1 - (x-2)^2$ on $\left[2, \frac{5}{2}\right]$ and $g = \frac{3}{2}(x-2)$ on $\left[\frac{5}{2}, 3\right]$.

Case $x \ge 3$: $[x-1, x] \subset [2, \infty)$ where f increases, so $g(x) = \frac{3}{2}(x-2)$.

Check the junctions. $x = 1$: $g'(1^-) = 2 - 2 = 0$ and $g'(1^+) = 0$. $x = 2$: $g'(2^-) = 0$ and $g'(2^+) = -2(x-2)\big|_{x=2} = 0$. $x = 3$: the formula $\frac{3}{2}(x-2)$ is the same on both sides. So g is differentiable at $1, 2, 3$.

$x = \frac{5}{2}$: $g'\left(\frac{5}{2}^-\right) = -2 \cdot \frac{1}{2} = -1$ and $g'\left(\frac{5}{2}^+\right) = \frac{3}{2}$. These differ, so g is not differentiable at $\frac{5}{2}$.

Inside each case g is a polynomial, so $S = \left\{\frac{5}{2}\right\}$ and the required sum is 2.50.

Final Answer: 2.50

Question 06

English

Let a and b be real numbers and let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} x^3, & x \text{ rational}, \\ ax + b, & x \text{ irrational}. \end{cases}$$

Suppose that f is continuous at $x = -1$ and that f is differentiable at some point $c \ne -1$. Then the value of $a + b$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि a तथा b वास्तविक संख्याएँ हैं तथा $f : \mathbb{R} \to \mathbb{R}$ इस प्रकार परिभाषित है कि x परिमेय होने पर $f(x) = x^3$ तथा x अपरिमेय होने पर $f(x) = ax + b$ है।

माना कि f, $x = -1$ पर सतत है तथा f किसी बिन्दु $c \ne -1$ पर अवकलनीय है। तब $a + b$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Every interval contains rationals and irrationals. If $x_0^3 \ne ax_0 + b$, then near x_0 the values of f approach two different numbers, so f is discontinuous at x_0. If $x_0^3 = ax_0 + b$, both pieces tend to the same value and f is continuous at x_0.

So f is continuous exactly at the real roots of $q(x) = x^3 - ax - b$. Continuity at -1 gives $q(-1) = 0$, i.e. $-1 + a - b = 0$.

Differentiability at c requires continuity, so $q(c) = 0$. Then for rational $x \to c$: $\dfrac{f(x) - f(c)}{x - c} = \dfrac{x^3 - c^3}{x - c} \to 3c^2$; for irrational $x \to c$: $\dfrac{ax + b - (ac + b)}{x - c} = a$. Both limits must agree, so $a = 3c^2$.

Conversely, if $q(c) = 0$ and $a = 3c^2$, both quotients tend to $3c^2$ and $f'(c) = 3c^2$.

$a = 3c^2$ means $q'(c) = 3c^2 - a = 0$, so c is a repeated root of q.

q is a cubic with roots -1 and the repeated root $c \neq -1$, so its roots are $-1, c, c$. The coefficient of x^2 in q is 0, so $-1 + 2c = 0$ and $c = \dfrac{1}{2}$.

Then $a = 3 \cdot \dfrac{1}{4} = \dfrac{3}{4}$ and $b = c^3 - ac = \dfrac{1}{8} - \dfrac{3}{8} = -\dfrac{1}{4}$. Check: $q(x) = x^3 - \frac{3}{4}x + \frac{1}{4} = \frac{1}{4}(x+1)(2x-1)^2$.

Hence $a + b = \dfrac{3}{4} - \dfrac{1}{4} = 0.50$.

Final Answer: 0.50

Question 07

English

Let $f : \mathbb{R} \to \mathbb{R}$ be a function which is continuous at $x = 0$ and satisfies

$$f(x) = f\left(\frac{x}{2}\right)\cos\left(\frac{x}{2}\right) \quad \text{for all } x \in \mathbb{R}.$$

If $f\left(\dfrac{\pi}{2}\right) = 140$, then the value of $f\left(\dfrac{\pi}{6}\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$ एक ऐसा फलन है जो $x = 0$ पर सतत है तथा सभी $x \in \mathbb{R}$ के लिए

$$f(x) = f\left(\frac{x}{2}\right)\cos\left(\frac{x}{2}\right)$$

को सन्तुष्ट करता है। यदि $f\left(\dfrac{\pi}{2}\right) = 140$ है, तो $f\left(\dfrac{\pi}{6}\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Replacing x by $\frac{x}{2}, \frac{x}{4}, \ldots$ in the relation and substituting back, for every positive integer n:

$$f(x) = f\left(\frac{x}{2^n}\right)\cos\frac{x}{2}\cos\frac{x}{4}\cdots\cos\frac{x}{2^n}.$$

Fix $x \neq 0$ and n large enough that $\sin\frac{x}{2^n} \neq 0$. Multiplying the product by $2^n \sin\frac{x}{2^n}$ and using $2\sin\theta\cos\theta = \sin 2\theta$ n times gives $\sin x$, so

$$\prod_{k=1}^{n}\cos\frac{x}{2^k} = \frac{\sin x}{2^n \sin\frac{x}{2^n}}.$$

As $n \to \infty$, $\frac{x}{2^n} \to 0$. By continuity of f at 0, $f\left(\frac{x}{2^n}\right) \to f(0)$. Also $2^n \sin\frac{x}{2^n} = x \cdot \dfrac{\sin(x/2^n)}{x/2^n} \to x$.

The left side $f(x)$ does not depend on n, so $f(x) = f(0)\dfrac{\sin x}{x}$ for all $x \neq 0$. (This f does satisfy the relation, since $\frac{\sin x}{x} = \frac{\sin(x/2)}{x/2}\cos\frac{x}{2}$.)

$f\left(\frac{\pi}{2}\right) = f(0) \cdot \dfrac{1}{\pi/2} = \dfrac{2f(0)}{\pi} = 140$, so $f(0) = 70\pi$.

$f\left(\frac{\pi}{6}\right) = 70\pi \cdot \dfrac{1/2}{\pi/6} = 70\pi \cdot \dfrac{3}{\pi} = 210$.

Final Answer: 210

Question 08

English

Let $a \in \mathbb{R}$ and let $f : [-1, 3] \setminus \{1\} \to \mathbb{R}$ be defined by

$$f(x) = \frac{x^3 - 3x^2 + ax + 5}{x - 1}.$$

Suppose that $\lim_{x\to 1} f(x)$ exists. Then the sum of all integers m for which the equation $f(x) = m$ has at least one solution in $[-1, 3] \setminus \{1\}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $a \in \mathbb{R}$ तथा $f : [-1, 3] \setminus \{1\} \to \mathbb{R}$ इस प्रकार परिभाषित है

$$f(x) = \frac{x^3 - 3x^2 + ax + 5}{x - 1}.$$

माना कि $\lim_{x\to 1} f(x)$ का अस्तित्व है। तब उन सभी पूर्णांकों m का योग, जिनके लिए समीकरण $f(x) = m$ का $[-1, 3] \setminus \{1\}$ में कम से कम एक हल है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

As $x \to 1$ the denominator tends to 0. If the numerator tended to a non-zero value, $|f(x)| \to \infty$ (an infinite discontinuity) and the limit would not exist. So the numerator vanishes at $x = 1$: $1 - 3 + a + 5 = 0$, giving $a = -3$.

Then $x^3 - 3x^2 - 3x + 5 = (x - 1)(x^2 - 2x - 5)$, so $f(x) = x^2 - 2x - 5 = (x - 1)^2 - 6$ for $x \in [-1, 3]$, $x \neq 1$, and $\lim_{x\to 1} f(x) = -6$. The discontinuity at 1 is removable, but 1 is not in the domain.

On $[-1, 1)$: f is continuous and strictly decreasing, $f(-1) = 4-6 = -2$, and $f(x) \to -6$ as $x \to 1^-$. By the intermediate value theorem on $[-1, t]$ for $t < 1$, f takes every value in $[f(t), -2]$; letting $t \to 1^-$, f takes every value in $(-6, -2]$.

On $(1, 3]$: similarly f increases from values near -6 to $f(3) = 4 - 6 = -2$, giving the same set $(-6, -2]$.

The value -6 would need $(x - 1)^2 = 0$, i.e. $x = 1$, which is excluded. So the range of f is $(-6, -2]$.

Integers in $(-6, -2]$: $-5, -4, -3, -2$. Their sum is -14.

Final Answer: -14

Question 09

English

For each positive integer n, let N_n denote the number of real solutions of the equation $e^x = x^n$. Then the value of $\sum_{n=1}^{10} N_n$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

प्रत्येक धनात्मक पूर्णांक n के लिए, माना कि N_n समीकरण $e^x = x^n$ के वास्तविक हलों की संख्या है। तब $\sum_{n=1}^{10} N_n$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

At $x = 0$: $e^0 = 1 \neq 0$, so $x = 0$ is not a solution.

Positive solutions. For $x > 0$ both sides are positive, and taking logarithms, $e^x = x^n \iff x = n \log_e x \iff \dfrac{\log_e x}{x} = \dfrac{1}{n}$.

Let $\phi(x) = \dfrac{\log_e x}{x}$ on $(0, \infty)$. $\phi'(x) = \dfrac{1 - \log_e x}{x^2}$, positive on $(0, e)$ and negative on (e, ∞). $\phi(x) \to -\infty$ as $x \to 0^+$, $\phi(e) = \dfrac{1}{e}$, and $\phi(x) \to 0^+$ as $x \to \infty$.

So for $0 < k < \frac{1}{e}$, $\phi(x) = k$ has exactly one solution in $(1, e)$ and one in (e, ∞) (IVT on each monotonic piece); for $k > \frac{1}{e}$ it has none.

$\frac{1}{n} < \frac{1}{e} \iff n > e \iff n \geq 3$: two positive solutions. For $n = 1, 2$: $\frac{1}{n} > \frac{1}{e}$, no positive solution.

Negative solutions, n odd: $x^n < 0 < e^x$, none.

Negative solutions, n even: let $h(x) = e^x - x^n$ on $(-\infty, 0]$. $h'(x) = e^x - nx^{n-1} > 0$ since $x^{n-1} < 0$. So h is strictly increasing; $h(x) \to -\infty$ as $x \to -\infty$ and $h(0) = 1 > 0$. By the IVT and monotonicity, h has exactly one zero in $(-\infty, 0)$.

Hence $N_1 = 0$, $N_2 = 1$, $N_n = 2$ for $n = 3, 5, 7, 9$ and $N_n = 3$ for $n = 4, 6, 8, 10$.

$$\sum_{n=1}^{10} N_n = 0 + 1 + (2 + 2 + 2 + 2) + (3 + 3 + 3 + 3) = 21.$$

Final Answer: 21

Question 10

English

Let a and b be real numbers and let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} \dfrac{ae^x - b - x}{x|x|}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Suppose that f has a jump discontinuity at $x = 0$, that is, $\lim_{x \to 0^-} f(x)$ and $\lim_{x \to 0^+} f(x)$ both exist as real numbers and are unequal. Then the value of $\lim_{x \to 0^-} f(x)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि a तथा b वास्तविक संख्याएँ हैं तथा $f : \mathbb{R} \to \mathbb{R}$ इस प्रकार परिभाषित है

$$f(x) = \begin{cases} \dfrac{ae^x - b - x}{x|x|}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

माना कि $x = 0$ पर f की उच्छेदी (jump) असांतत्यता है, अर्थात $\lim_{x \to 0^-} f(x)$ तथा $\lim_{x \to 0^+} f(x)$ दोनों वास्तविक संख्याओं के रूप में विद्यमान हैं तथा असमान हैं। तब $\lim_{x \to 0^-} f(x)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Expand the numerator: $ae^x - b - x = (a - b) + (a - 1)x + \dfrac{a}{2}x^2 + O(x^3)$. The denominator is $x|x|$, with $|x|x|| = x^2$; it equals x^2 for $x > 0$ and $-x^2$ for $x < 0$.

If $a \neq b$, the numerator tends to $a - b \neq 0$ while the denominator tends to 0, so $|f(x)| \to \infty$: an infinite discontinuity, not a jump. Hence $a = b$.

If $a = b$ but $a \neq 1$, then $f(x) = \dfrac{(a-1)x + O(x^2)}{x|x|} = \dfrac{a-1}{|x|} + O(1)$, which is unbounded near 0. Hence $a = 1$, and so $b = 1$.

With $a = b = 1$: $\dfrac{e^x - 1 - x}{x^2} \to \dfrac{1}{2}$ as $x \to 0$ (the x^2 coefficient of e^x).

For $x > 0$: $f(x) = \dfrac{e^x - 1 - x}{x^2} \to \dfrac{1}{2}$. For $x < 0$: $f(x) = -\dfrac{e^x - 1 - x}{x^2} \to -\dfrac{1}{2}$.

The one-sided limits are $-\frac{1}{2}$ and $\frac{1}{2}$, both finite and unequal, so this is indeed a jump discontinuity. Therefore $\lim_{x \to 0^-} f(x) = -0.50$.

Final Answer: -0.50

Question 11

English

Let a, b, c, d be real numbers and let $f : \mathbb{R} \to \mathbb{R}$ be the function such that $f(x) = a + bx + cx^2 + dx^3$ for $x < 0$ and, for $x \geq 0$, $f(x)$ is the unique real number y satisfying $y + 6y^3 = x$. If f is three times differentiable at $x = 0$, then the value of $f\left(-\dfrac{1}{2}\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि a, b, c, d वास्तविक संख्याएँ हैं तथा $f : \mathbb{R} \to \mathbb{R}$ एक ऐसा फलन है कि $x < 0$ के लिए $f(x) = a + bx + cx^2 + dx^3$ है तथा $x \geq 0$ के लिए $f(x)$ वह अद्वितीय वास्तविक संख्या y है जो $y + 6y^3 = x$ को सन्तुष्ट करती है। यदि f, $x = 0$ पर तीन बार अवकलनीय है, तो $f\left(-\dfrac{1}{2}\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$u(y) = y + 6y^3$ is strictly increasing and onto $\mathbb{R}$, so for $x \geq 0$ the value $y = f(x)$ exists, is unique, and $y \geq 0$; at $x = 0$, $y = 0$. By the inverse function rule y is differentiable any number of times for $x \geq 0$.

Differentiate $y + 6y^3 = x$: $y'(1 + 18y^2) = 1$. At $x = 0$: $y'(0) = 1$.

Differentiate again: $y''(1 + 18y^2) + 36y\,(y')^2 = 0$. At $x = 0$, $y = 0$: $y''(0) = 0$.

Differentiate again: $y'''(1+18y^2)+36y\,y'y''+36\left[(y')^3 + 2y\,y'y''\right] = 0$, i.e. $y'''(1+18y^2)+108y\,y'y''+36(y')^3 = 0$. At $x = 0$: $y'''(0) = -36$.

These are the right-hand values $f(0) = 0$, $f'(0^+) = 1$, $f''(0^+) = 0$, $f'''(0^+) = -36$.

For $x < 0$, $f(x) = a + bx + cx^2 + dx^3$ gives left-hand values a, b, $2c$, $6d$ at 0.

f three times differentiable at 0 requires continuity of f, f', f'' at 0 and equal one-sided third derivatives: $a = 0$, $b = 1$, $2c = 0$, $6d = -36$. So $c = 0$, $d = -6$.

Hence $f(x) = x - 6x^3$ for $x < 0$ and $f\left(-\dfrac{1}{2}\right) = -\dfrac{1}{2} - 6\left(-\dfrac{1}{8}\right) = -\dfrac{1}{2} + \dfrac{3}{4} = \dfrac{1}{4} = 0.25$.

Final Answer: 0.25

Question 12

English

For positive integers m and n, let $f_{m,n} : \mathbb{R} \to \mathbb{R}$ be defined by

$$f_{m,n}(x) = \begin{cases} x^m \sin\left(\dfrac{1}{x^n}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Let N be the number of ordered pairs (m, n) with $m, n \in \{1, 2, \ldots, 20\}$ for which $f_{m,n}$ is differentiable at $x = 0$ and $f'_{m,n}$ is unbounded on $(-\delta, \delta) \setminus \{0\}$ for every $\delta > 0$. Then the value of N is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

धनात्मक पूर्णांकों m तथा n के लिए, माना कि $f_{m,n} : \mathbb{R} \to \mathbb{R}$ इस प्रकार परिभाषित है

$$f_{m,n}(x) = \begin{cases} x^m \sin\left(\dfrac{1}{x^n}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

माना कि N उन क्रमित युग्मों (m, n), जहाँ $m, n \in \{1, 2, \ldots, 20\}$, की संख्या है जिनके लिए $f_{m,n}$, $x = 0$ पर अवकलनीय है तथा प्रत्येक $\delta > 0$ के लिए $f'_{m,n}$, $(-\delta, \delta) \setminus \{0\}$ पर अपरिबद्ध है। तब N का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Differentiability at 0: $\dfrac{f_{m,n}(x) - 0}{x} = x^{m-1} \sin\left(\dfrac{1}{x^n}\right)$. If $m \geq 2$, $\left|x^{m-1} \sin \frac{1}{x^n}\right| \leq |x|^{m-1} \to 0$, so $f'_{m,n}(0) = 0$. If $m = 1$, the quotient is $\sin \frac{1}{x^n}$, which takes the values 1 and -1 on sequences tending to 0, so it has no limit. Hence $f_{m,n}$ is differentiable at 0 iff $m \geq 2$.

For $x \neq 0$: $f'_{m,n}(x) = mx^{m-1} \sin\left(\dfrac{1}{x^n}\right) + x^m \cos\left(\dfrac{1}{x^n}\right) \cdot \left(-\dfrac{n}{x^{n+1}}\right) = mx^{m-1} \sin \dfrac{1}{x^n} - n\, x^{m-n-1} \cos \dfrac{1}{x^n}$.

With $m \geq 2$ the first term is bounded by $m|x|^{m-1} \leq m$ on $(-1, 1)$.

If $m \geq n + 1$, the second term is bounded by $n|x|^{m-n-1} \leq n$ on $(-1, 1)$, so f' is bounded near 0.

If $m \leq n$, take $x_k = (2k\pi)^{-1/n} \to 0^+$. Then $\sin \frac{1}{x_k^n} = 0$, $\cos \frac{1}{x_k^n} = 1$ and $|f'(x_k)| = n\, x_k^{m-n-1} \to \infty$ since $m - n - 1 \leq -1$. So f' is unbounded on every $(-\delta, \delta) \setminus \{0\}$.

So the required pairs are those with $2 \leq m \leq n$, $m, n \leq 20$.

For a fixed $n \in \{2, \ldots, 20\}$, m takes the $n - 1$ values $2, \ldots, n$; for $n = 1$ there is none.

$$N = \sum_{n=2}^{20} (n - 1) = \frac{19 \cdot 20}{2} = 190.$$

Final Answer: 190

Question 13

English

Let $f : (0, \infty) \to \mathbb{R}$ be defined by $f(x) = \left|\cos\left(\dfrac{7\pi}{2x}\right)\right|$. For $0 < a < 1$, let $N(a)$ denote the number of points of the interval $(a, 1)$ at which f is not differentiable. Then the value of $\lim\limits_{a \to 0^+} a\, N(a)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : (0, \infty) \to \mathbb{R}$, $f(x) = \left|\cos\left(\frac{7\pi}{2x}\right)\right|$ द्वारा परिभाषित है। $0 < a < 1$ के लिए, माना कि $N(a)$ अन्तराल $(a, 1)$ के उन बिन्दुओं की संख्या है जिन पर f अवकलनीय नहीं है। तब $\lim\limits_{a \to 0^+} a\, N(a)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $g(x) = \cos\left(\frac{7\pi}{2x}\right)$ on $(0, \infty)$; g is differentiable and $f = |g|$.

Where $g \neq 0$, $f = \pm g$ near the point and is differentiable. At a zero x_0 of g with $g'(x_0) \neq 0$, g changes sign and f has one-sided derivatives $-|g'(x_0)|$ and $|g'(x_0)|$, so f is not differentiable there.

Zeros of g: $\frac{7\pi}{2x} = \left(k + \frac{1}{2}\right)\pi$, i.e. $x = \frac{7}{2k+1}$, $k = 0, 1, 2, \ldots$

$g'(x) = \frac{7\pi}{2x^2}\sin\left(\frac{7\pi}{2x}\right)$, and at a zero of the cosine the sine is ± 1, so $g' \neq 0$. Hence f is non-differentiable exactly at the points $\frac{7}{2k+1}$.

Such a point lies in $(a, 1)$ iff $2k + 1 > 7$ and $2k + 1 < \frac{7}{a}$, i.e. $k \geq 4$ and $k < \frac{7}{2a} - \frac{1}{2}$. ($k = 3$ gives $x = 1$, which is excluded.)

$N(a)$ is the number of integers k with $4 \leq k < \frac{7}{2a} - \frac{1}{2}$. If M is the largest such k, then $N(a) = M - 3$ and $\frac{7}{2a} - \frac{3}{2} \leq M < \frac{7}{2a} - \frac{1}{2}$, hence $\frac{7}{2a} - \frac{9}{2} \leq N(a) < \frac{7}{2a} - \frac{7}{2}$ (for small a).

Multiplying by $a > 0$: $\frac{7}{2} - \frac{9a}{2} \leq aN(a) < \frac{7}{2} - \frac{7a}{2}$.

Both bounds tend to $\frac{7}{2}$ as $a \to 0^+$, so by the sandwich theorem $\lim\limits_{a \to 0^+} aN(a) = \frac{7}{2} = 3.50$.

Final Answer: 3.50

Question 14

English

Let a and b be real numbers. Let $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^3 + ax^2 + bx$ and

$$g(x) = \begin{cases} x - 1, & x < 0, \\ 4x + 1, & x \geq 0. \end{cases}$$

If the composite function $f \circ g$ is differentiable at $x = 0$, then the value of $(f \circ g)'(0)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि a तथा b वास्तविक संख्याएँ हैं। माना कि $f : \mathbb{R} \to \mathbb{R}$ तथा $g : \mathbb{R} \to \mathbb{R}$, $f(x) = x^3 + ax^2 + bx$ तथा

$$g(x) = \begin{cases} x - 1, & x < 0, \\ 4x + 1, & x \geq 0. \end{cases}$$

द्वारा परिभाषित हैं। यदि संयुक्त फलन $f \circ g$, $x = 0$ पर अवकलनीय है, तो $(f \circ g)'(0)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

g is not continuous at 0: $\lim_{x\to 0^-} g(x) = -1$ while $g(0) = 1$. So the chain rule cannot be applied at 0, and $h = f \circ g$ must be examined from each side.

For $x < 0$: $h(x) = f(x-1)$; for $x \geq 0$: $h(x) = f(4x+1)$. Both are polynomials.

Differentiability implies continuity, so $\lim_{x\to 0^-} h(x) = h(0)$, i.e. $f(-1) = f(1)$.

$f(1) = 1 + a + b$ and $f(-1) = -1 + a - b$. Equating: $2 + 2b = 0$, so $b = -1$.

Now $f'(x) = 3x^2 + 2ax - 1$, so $f'(-1) = 2 - 2a$ and $f'(1) = 2 + 2a$.

Since $h(0^-) = h(0)$, the left derivative is $\lim_{x\to 0^-} \dfrac{f(x-1) - f(-1)}{x} = f'(-1) = 2 - 2a$. The right derivative is $\lim_{x\to 0^+} \dfrac{f(4x+1) - f(1)}{x} = 4f'(1) = 8 + 8a$.

Equal one-sided derivatives: $2 - 2a = 8 + 8a$, so $a = -\dfrac{3}{5}$.

Then $(f \circ g)'(0) = 2 - 2a = 2 + \dfrac{6}{5} = \dfrac{16}{5} = 3.20$ (and $4f'(1) = 4 \cdot \dfrac{4}{5} = \dfrac{16}{5}$ agrees).

Final Answer: 3.20